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Metformin in T2DM: neurocognitive mechanisms and precision pharmacotherapy

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Abstract: Type 2 diabetes mellitus (T2DM) is a complex metabolic disorder that significantly predisposes individuals to delirium and dementia through multifaceted neurobiological pathways. The essence of this neurocognitive decline involves mechanisms such as central insulin resistance, chronic low-grade inflammation, and mitochondrial dysfunction. While metformin remains the cornerstone of T2DM management, its impact on the central nervous system exhibits a “double-edged sword” nature, balancing intrinsic neuroprotective properties against the potential neurotoxicity associated with vitamin B12 deficiency. This review aims to systematically synthesize epidemiological and clinical evidence linking metformin to neurocognitive outcomes, contrasting its efficacy with newer glucose-lowering agents such as GLP-1 receptor agonists and SGLT2 inhibitors. In addition, it sheds light on the reciprocal connectivity between systemic metabolic regulation and direct CNS modulation, specifically elucidating AMPK activation, the autophagy–lysosome axis, and the gut–brain and liver–brain axes. We review these molecular mechanisms to delineate the delicate trade-off between neuroprotection and risk, providing a framework for precision pharmacotherapy and biomarker-guided stratification in high-risk T2DM populations.

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1 Introduction

Metformin, a derivative of biguanide, has a discovery history rooted in the traditional European herbal plant known as the goat bean. As early as 1918, scientists identified that guanidine compounds abundant in this plant had the ability to lower blood sugar levels. The subsequent chemical synthesis of the drug led to the creation of metformin, although its clinical significance was not fully recognized until the mid-20th century (Bailey 2017). In 1957, French physician Jean Sterne first reported the use of metformin in the treatment of diabetes, marking a pivotal moment in its modern pharmacological development (Sterne 1957). After decades of clinical validation – most notably through the landmark UK Prospective Diabetes Study (UKPDS), which established its long-term cardiovascular benefits – metformin was officially introduced to the US market in 1995. It has since become the preferred first-line oral medication for the treatment of type 2 diabetes (T2DM) worldwide (Bailey 2017; UK Prospective Diabetes Study Group 1998).

Although metformin was initially utilized solely as an antidiabetic medication, ongoing research has revealed that its pharmacological effects extend far beyond mere blood sugar control, exhibiting extensive “pleiotropy.” Mechanistic studies have demonstrated that metformin effectively activates AMP-activated protein kinase (AMPK), a key regulator of cellular energy metabolism, which is considered the primary mechanism behind its diverse biological effects (Zhou et al. 2001). Building on this understanding, the potential applications of metformin in nondiabetic areas have continually emerged. For example, early clinical studies found that metformin significantly improved hyperinsulinemia and hyperandrogenism in patients with polycystic ovary syndrome (PCOS) (Velazquez et al. 1994). Furthermore, epidemiological evidence indicates that metformin treatment is strongly associated with a reduced risk of cancer among diabetic patients, suggesting its potential antitumor properties (Evans et al. 2005). Currently, multiple recent articles have further confirmed its efficacy in specific tumor types, including lung, breast, ovarian, and endometrial cancers (Dutta et al. 2023; Galal et al. 2024; Sirtori et al. 2024; Smith et al. 2025).

In recent years, the potential neuroprotective effects of metformin have garnered considerable attention within the academic community. A groundbreaking study by Chen et al. (2009) was the first to report that metformin could reverse cognitive deficits and reduce β-amyloid deposition in Alzheimer's disease (AD) model mice by activating the AMPK pathway, marking the inception of research into its neuroprotective properties. Since then, its therapeutic potential has been validated across various neurological disease models.

In models of autoimmune diseases, metformin has demonstrated the ability to alleviate the severity of multiple sclerosis by limiting inflammatory cell infiltration (Nath et al. 2009). Furthermore, in Huntington's disease research, metformin has been shown to protect neurons by enhancing autophagy, and clinical observations suggest an association with improved cognitive status in patients (Hervás et al. 2017; Vázquez-Manrique et al. 2016). Currently, multiple recent studies continue to substantiate these neuroprotective properties. Evidence highlights metformin's capacity to cross the blood–brain barrier and stimulate neurogenesis, reinforcing its enduring therapeutic potential across a broader spectrum of central and peripheral nervous system disorders, including Alzheimer's disease, Parkinson's disease, and others (Campagnoli et al. 2025; Dutta et al. 2023; Li et al. 2024; Nowell et al. 2023).

Notably, this potential has successfully bridged the translational gap. A randomized controlled trial (RCT) involving patients with severe traumatic brain injury (TBI) revealed that early administration of metformin was safe, did not lead to lactic acidosis, and significantly reduced serum brain injury markers, such as S100b. However, the study also highlighted pharmacokinetic challenges related to delayed drug absorption in critically ill patients (Taheri et al. 2019).

However, the role of metformin in the nervous system is not always straightforward; its effects are highly complex and context-dependent. Li et al. (2010) found that in an ischemic stroke model, while long-term preventive administration of metformin exhibited neuroprotective effects, acute administration during the critical phase actually exacerbated brain damage. This highlights its "double-edged sword" effect and underscores the importance of the timing of administration (Li et al. 2010).

This complexity is particularly significant in its primary area of application – type 2 diabetes–related cognitive impairment (T2DM-related cognitive impairment). Epidemiological studies have long established that diabetes is an independent risk factor for dementia and Alzheimer's disease (AD) (Ott et al. 1999; Zilliox et al. 2016), with poor blood sugar control accelerating cognitive decline (Dove et al. 2021). Due to the substantial overlap in pathological mechanisms between AD and diabetes, this has led some researchers to propose the conceptual framework of "type 3 diabetes" to describe the phenomenon of brain insulin resistance in AD. It is important to note that this term is not a formal clinical diagnosis but rather a controversial hypothesis used to highlight the shared metabolic dysregulation pathways between the two conditions (de la Monte and Wands 2008).

Additionally, certain mechanistic studies have revealed that extracellular vesicles (EVs) from peripheral adipose tissue and abnormal polarization of microglia are critical links leading to cognitive impairment (Tian et al. 2024; Wang et al. 2022). Given this context, this article aims to systematically review the role of metformin in neurocognitive function related to T2DM, deeply analyze its complex molecular mechanisms and potential risks, and explore its future applications in the era of precision medicine.

2 Metformin and T2DM brain health: integrating clinical evidence with neurobiological mechanisms

2.1 Epidemiological and clinical evidence: from T2DM to delirium and dementia

2.1.1 Underlying risk of T2DM and delirium/dementia

Numerous cohort studies and systematic reviews have demonstrated robust epidemiological associations between T2DM and several neurodegenerative diseases, including Alzheimer's disease (AD), vascular dementia, and Parkinson's disease (PD) (Madhusudhanan et al. 2020; Szablewski 2025). Building on this evidence, a growing number of studies are investigating the neuroprotective or anti-dementia potential of hypoglycemic agents such as metformin (Cui et al. 2024; Du et al. 2022; Kruczkowska et al. 2025). A review by Szablewski (2025) indicates that both type 1 and type 2 diabetes are associated with cognitive decline and multiple neurodegenerative diseases, with T2DM promoting pathological progression in the brain through impaired insulin signaling, mitochondrial damage, and inflammatory responses. Within a more nuanced pathological framework, AD is sometimes referred to under the conceptual label of "type 3 diabetes." While this remains a debated hypothesis rather than a validated classification, it serves to underscore the intertwined nature of central insulin resistance with amyloid and tau pathology (Kciuk et al. 2024).

With respect to acute outcomes, systematic reviews and meta-analyses of diabetes and delirium risk have shown an overall increased incidence of delirium among patients with diabetes, particularly in postoperative and intensive care settings. This suggests that metabolic dysregulation and insulin resistance may erode the brain's "cognitive reserve" against acute stressors (Komici et al. 2025). Furthermore, T2DM is associated with a greater burden of cerebral small vessel disease and poststroke cognitive decline, with these chronic vascular pathologies constituting key substrates for dementia development, particularly mixed dementia.

2.1.2 Cohort studies and meta-analyses of metformin and dementia/delirium

Several early cohort studies have suggested that metformin use is associated with an overall reduction in the risk of dementia or other neurodegenerative diseases (Chin-Hsiao 2019; Zhang et al. 2022). Using Taiwanese nationwide databases, Chin-Hsiao (2019) conducted a retrospective cohort study and observed an inverse association between metformin use and dementia risk, although this effect was not entirely consistent across different age groups and durations of treatment.

More recent large-scale studies have further reinforced the evidence for a dose–response relationship. In a cohort of older adults with T2DM, Sun et al. (2024) used time-dependent Cox models and found that metformin use was associated with a significantly lower risk of dementia, within the safe therapeutic dose range, with both higher daily and cumulative doses corresponding to progressively lower hazard ratios. In another study, Sun et al. (2025a) reported that metformin use was likewise associated with a significantly reduced risk of delirium in older patients with T2DM, again displaying a clear dose–response pattern. Taken together, these findings suggest that, in older adults, adequate and sustained metformin therapy may confer protective effects on both acute and chronic neurocognitive outcomes.

At the meta-analytic level, Campbell et al. (2018) pooled 14 studies and reported that metformin use was associated with a hazard ratio of approximately 0.76 for incident dementia (95 % CI 0.39–0.88) and an odds ratio of about 0.55 for prevalent cognitive impairment. Zhang et al. (2022) combined 12 population-based cohorts and found that metformin use was associated with an overall reduced risk of neurodegenerative diseases (RR 0.77), with substantially greater benefit observed among long-term users (≥4 years; RR 0.29). In a meta-analysis of 20 cohorts including more than 3.4 million patients, Tang et al. (2025) confirmed that, among individuals with T2DM, metformin use was significantly inversely associated with all-cause dementia (HR 0.76) and was superior to sulfonylureas (HR 0.88), although the protective effect was not evident in newly diagnosed T2DM.

By contrast, meta-analytic findings specific to AD are more nuanced. Luo et al. (2022) included 10 observational studies and found no significant overall association between metformin use and AD risk but observed an increased risk of AD among metformin users in Asian populations (OR 1.71), suggesting that ethnicity, baseline risk, and comorbid factors may influence the direction of the effect. Such heterogeneity has also been repeatedly reported in several studies of dementia not specific to AD (Adem et al. 2024; Hui et al. 2025). Although the underlying reasons remain unclear, this heterogeneity may be associated with factors such as genetics, diet, comorbidity patterns, and prescribing habits, warranting further investigation in future studies.

With respect to delirium, a recent multinational cohort study of older adults with T2DM showed that, compared with second-line glucose-lowering regimens, first-line or continuous metformin use was associated with a significantly reduced risk of delirium, and that this protective effect became more pronounced with longer treatment duration and higher doses (Sun et al. 2025b).

2.1.3 Real-world comparisons with SGLT2 inhibitors, GLP-1 receptor agonists, DPP-4 inhibitors, etc.

In the context of the rapid expansion of novel glucose-lowering therapies, merely asking whether metformin has an effect is no longer sufficient to address clinical practice; the key question now concerns its relative advantage for neurocognitive outcomes compared with agents such as sodium–glucose cotransporter-2 inhibitors (SGLT2is), glucagon-like peptide-1 receptor agonists (GLP-1RAs), and dipeptidyl peptidase-4 inhibitors (DPP-4is) (Table 1).

Hui et al. (2025) found that GLP-1RAs significantly reduced the risk of subsequent dementia in both RCTs and case–control studies, while thiazolidinediones also demonstrated a consistent protective effect. By contrast, metformin, sulfonylureas, DPP-4is, and insulin did not confer a statistically significant overall reduction in dementia risk in the pooled analyses. The apparent advantage of GLP-1RAs may be attributable to their broad neuroprotective actions, including attenuation of neuroinflammation, promotion of neuronal

Table 1: Comparative efficacy and mechanisms of glucose-lowering agents on neurocognitive outcomes in T2DM.

Drug class	Primary neuroprotective mechanism	Clinical evidence for dementia risk	Potential limitations/risks
Metformin	AMPK activation: enhances autophagy, inhibits NLRP3 inflammasome, improves gut–brain axis (*A. muciniphila*).	Associated with reduced risk in long-term users (Tang et al. 2025); effective in early MCI.	Vitamin B12 deficiency: risk of neuropathy; potential mitochondrial inhibition in acute stress; heterogeneous effects in Asian cohorts.
GLP-1 RAs	Broad neuroprotection: antineuroinflammation, promotes neuronal growth, improves central insulin signaling.	Consistent reduction in dementia risk; potentially superior to metformin in head-to-head comparisons (Sun et al. 2025d).	High cost; gastrointestinal side effects; injectable forms may reduce adherence in some elderly patients.
SGLT2 inhibitors	Vascular & metabolic: improves endothelial function, reduces oxidative stress, lowers vascular burden	May be superior to metformin in patients with high cardiovascular/renal risk (Sun et al. 2025c).	Risk of genitourinary infections; less direct evidence for central neuronal modulation compared to GLP-1 RAs.
DPP-4 inhibitors	Increases endogenous GLP-1 levels.	Most meta-analyses show no significant benefit compared to metformin or GLP-1 RAs (Hui et al. 2025).	Generally safe but lacks strong evidence for robust neurocognitive protection.

AMPK, AMP-activated protein kinase; DPP-4, dipeptidyl peptidase-4; GLP-1 RAs, glucagon-like peptide-1 receptor agonists; MCI, mild cognitive impairment; NLRP3, NOD-like receptor family pyrin domain containing 3; SGLT2, sodium-glucose cotransporter-2; T2DM, type 2 diabetes mellitus.

growth, and improvement of central metabolic function (Zhao et al. 2021). A study published in JAMA Neurology further reported that, when glucose-lowering agents with established cardiovascular benefits were evaluated as a group, there was no significant overall benefit for dementia or cognitive impairment. However, in class-specific analyses, GLP-1RAs were associated with a significant reduction in all-cause dementia, whereas SGLT2is did not show a clear effect (Seminer et al. 2025).

For head-to-head, real-world comparisons among glucose-lowering drugs, the data are particularly informative. Milori et al. (2025) found that SGLT2i users had a lower risk of dementia than DPP-4i users, whereas no significant difference was observed in comparison with GLP-1RAs. However, both comparisons were characterized by substantial heterogeneity. In a network meta-analysis synthesizing multiple studies in this field, the risks of dementia and AD associated with different glucose-lowering agents were compared within a Bayesian framework, and the overall findings indicated that metformin and SGLT2is were associated with a lower risk of dementia than other antidiabetic drugs (Sunwoo et al. 2024).

For studies using metformin as the reference, Sun et al. (2025c) compared SGLT2 inhibitors (SGLT2is) with metformin as first-line therapy for T2DM in relation to dementia prevention and found that SGLT2is appeared to be slightly more effective than metformin in reducing dementia risk, particularly among older patients with high cardiovascular or renal risk. Another retrospective cohort study drawing on multiple global databases directly compared GLP-1 receptor agonists (GLP-1RAs) with metformin as first-line monotherapy and showed that GLP-1RAs were associated with lower risks of all-cause dementia, AD, and nonvascular dementia than metformin, with benefits especially pronounced in older women (Sun et al. 2025d). However, not all high-quality studies have demonstrated a clear advantage of GLP-1RAs. A target trial emulation in older adults with diabetes found no definitive evidence of a significant difference in overall dementia incidence between users of GLP-1RAs and dipeptidyl peptidase-4 inhibitors (DPP-4is) (Inoue et al. 2025). Regarding combination therapy, a Taiwan-based cohort study showed that, among patients with newly diagnosed T2DM, treatment with metformin plus a DPP-4i or a thiazolidinedione (TZD) was associated with a lower risk of dementia than metformin combined with a sulfonylurea or a glinide (Hsieh et al. 2025).

2.1.4 "Contradictions" between MR findings and clinical data: the role of indication bias, dosage, and comorbidities

From a causal inference perspective, Mendelian randomization (MR) is a statistical approach that uses genetic variants as instruments to infer causal relationships between risk factors and outcomes, thereby mitigating potential confounding and providing an important complement to our understanding of the interrelations among diabetes, dementia, and glucose-lowering therapies. A two-sample MR analysis showed that genetic liability to T2DM was positively associated with AD risk, whereas genetic proxies for metformin targets (such as AMPK activation) were inversely associated with AD risk, indicating that diabetes may increase the risk of AD, while metformin may exert a causal effect in the opposite direction (Liu et al. 2025). Another study used AMPK activation as an instrumental variable and found that it was associated with improved 3-month functional outcomes after ischemic stroke, offering genetic support for the hypothesis that metformin, via AMPK targeting, could improve prognosis after cerebrovascular events (Wang et al. 2023).

However, MR findings are not always aligned with those from real-world observational studies. For example, some MR analyses have failed to confirm a causal relationship between T1DM and AD and have even suggested that genetic liability to T1DM is inversely associated with susceptibility to Parkinson's disease (PD) (Geng et al. 2024). At the drug-specific level, most MR studies of individual glucose-lowering agents use drug targets as instrumental variables, making it difficult to fully capture the "treatment exposure package" defined by clinical practice, including prescribed dose, route of administration, and concomitant medications (Liu et al. 2025).

The apparent "discrepancies" between MR studies and clinical data largely reflect the following issues:

1) Indication bias: Metformin is usually prescribed in the earlier stages of disease, in patients who exhibit a prominent metabolic syndrome phenotype and do not yet have severe renal impairment, whereas those with advanced T2DM and multiple organ complications are more likely to receive insulin or other agents. As a result, the "metformin group" intrinsically tends to have a better prognosis and greater cognitive reserve (Hui et al. 2025; Seminer et al. 2025; Sun et al. 2025c).
2) Differences in dose and duration: MR reflects lifelong or long-term "genetic exposure," whereas in clinical practice, metformin is often initiated in midlife or even later and may be intermittently discontinued because of adverse effects or impaired renal function (Liu et al. 2025; Sun et al. 2024; Tang et al. 2025).
3) Comorbidity burden and vitamin B12 deficiency: Long-term metformin use is clearly associated with vitamin B12 deficiency and related peripheral and central nervous system damage (Sayedali et al. 2023). Among malnourished, very old, and polymedicated individuals, this "adverse effect" may offset or even reverse part of its neuroprotective benefits (Al-kuraishy et al. 2023; Campbell et al. 2018; Luo et al. 2022).
4) Differences in endpoint definition: MR studies mostly use clinically diagnosed AD or "GWAS-defined AD" as the outcome, whereas observational studies often adopt "all-cause dementia," primary diagnostic codes, or declines in cognitive test scores. Differences in the underlying pathological spectrum and diagnostic accuracy of these endpoints may lead to inconsistencies in both the direction and magnitude of effect estimates (Hui et al. 2025; Seminer et al. 2025; Tang et al. 2025).

In summary, T2DM markedly increases the risk of both delirium and dementia, with multiple mechanisms – such as insulin resistance, chronic low-grade inflammation, and vascular pathology – playing key roles in this process. Accumulating epidemiological and clinical evidence suggests that metformin, a widely used glucose-lowering agent, may confer beneficial effects on neurocognitive function through several pathways. However, findings regarding the protective effects of metformin remain heterogeneous. In the following sections, we will examine in greater detail the mechanisms of action of metformin, with the aim of further elucidating the complexity of neurocognitive outcomes in patients with T2DM and identifying potential therapeutic strategies.

2.2 Mechanistic insights: from systemic regulation to cellular modulation

2.2.1 Indirect neuroprotection via systemic metabolic improvements and the gut–brain axis

At the pathological level, the relationship between T2DM and AD is largely linked by "brain insulin resistance." A PET study by Rebelos et al. (2021) showed that under standardized hyperinsulinemic–euglycemic clamp conditions, individuals with greater insulin resistance paradoxically exhibited higher cerebral glucose uptake, suggesting that cerebral metabolic patterns undergo adaptive reorganization in the setting of insulin resistance. Animal and cellular models further demonstrate that impaired insulin signaling can promote tau phosphorylation and Aβ production via the PI3K–Akt–GSK3β pathway (phosphoinositide 3-kinase/protein kinase B/glycogen synthase kinase-3β), thereby disrupting synaptic plasticity and memory formation (Kciuk et al. 2024; Madhusudhanan et al. 2020). Recent clinical evidence in patients with AD provides additional support: one analysis found that, among various antidiabetic agents, metformin was the most effective in reducing cerebrospinal fluid Aβ levels (Cai et al. 2025).

Within this framework, metformin, as a peripheral insulin sensitizer, reduces hepatic glucose output and improves peripheral glucose utilization through AMPK activation, thereby indirectly alleviating cerebral "glucotoxicity" and the burden of brain insulin resistance. As a systemic metabolic modulator, metformin exerts neuroprotective effects not only through direct actions within the central nervous system but also by regulating the gut–brain and liver–brain axes (Figure 1). Studies have shown that metformin can markedly remodel the gut microbiota, specifically increasing the abundance of beneficial bacteria

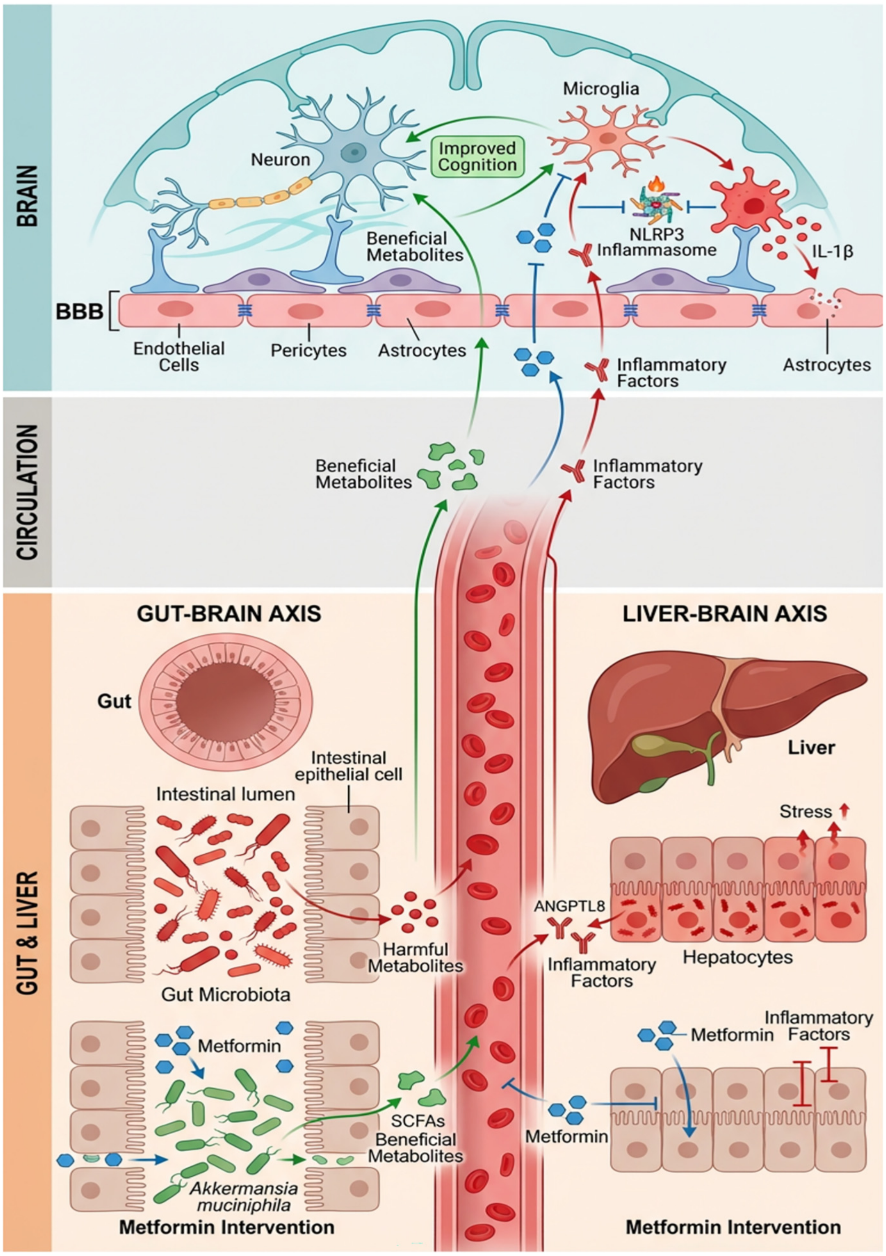

Figure 1: Systemic neuroprotective mechanisms of metformin via the gut–brain and liver–brain axes. The schematic illustrates the indirect neuroprotective effects of metformin mediated by peripheral organ systems. In the gut–brain axis (bottom left), metformin intervention remodels the gut microbiota, specifically increasing the abundance of beneficial bacteria such as *Akkermansia muciniphila*. This shift enhances the production of short-chain fatty acids (SCFAs) and other beneficial metabolites, which enter the circulation and cross the blood–brain barrier (BBB) to improve cognition. In the liver–brain axis (bottom right), metabolic stress in hepatocytes triggers the release of inflammatory factors (e.g., ANGPTL8). Metformin inhibits the secretion of these hepatic inflammatory mediators, thereby reducing the systemic inflammatory burden. Consequently, the reduced influx of harmful metabolites and inflammatory factors across the BBB attenuates microglial activation and suppresses the NLRP3 inflammasome pathway in the brain, reducing the release of IL-1β and preserving neuronal function.

such as *Akkermansia muciniphila*. More importantly, these microbiota changes are accompanied by elevated levels of plasma metabolites, which in turn improve memory and executive function via gut–brain axis mechanisms (Rosell-Díaz et al. 2024).

At the level of the liver–brain axis, liver-derived factors such as ANGPTL8 are thought to mediate the transmission of peripheral inflammatory signals to central pyroptosis through the PirB/NLRP3 pathway. Although clinical studies directly examining metformin in this pathway are still limited, metformin has been shown to inhibit activation of the NLRP3 inflammasome, suggesting that it may interfere with liver-derived inflammatory mediators and thereby disrupt the vicious cycle of "hyperglycemia–hepatic stress–neuroinflammation" (Meng et al. 2024; Wei et al. 2025). This systemic regulatory mechanism provides important clues for "precision medicine": in T2DM patients with severe nonalcoholic fatty liver disease (NAFLD) or gut dysbiosis, metformin may delay cognitive decline more effectively by "peripheral clearance" than by central interventions alone.

2.2.2 Direct CNS effects: modulating synapses, inflammation, and the epigenetic clock

Some studies have indicated that metformin can cross the blood–brain barrier and directly modulate AMPK activity and mitochondrial function in neurons and glial cells (Campagnoli et al. 2025; Koenig et al. 2017; Papini et al. 2024). Transport of metformin across the blood–brain barrier depends primarily on the Oct3 and PMAT transporters; fluctuations in their expression and functional alterations under pathological conditions (such as hypoxia and hypoglycemia) may constitute the pharmacokinetic basis for the pronounced heterogeneity of metformin's neuroprotective effects observed in clinical studies (Sharma et al. 2023). In a small randomized crossover trial in patients with AD, short-term metformin administration resulted in stable, detectable concentrations in the cerebrospinal fluid and was associated with improvements in executive function and a trend toward increased prefrontal blood flow (Koenig et al. 2017). Furthermore, these targeted neuroprotective effects and anti-inflammatory mechanisms have been robustly corroborated by recent clinical evidence in acute settings. A recent observational study by Akiyama et al. (2024) demonstrated that patients with type 2 diabetes pretreated with metformin exhibited significantly lower serum levels of the proinflammatory cytokine IL-6 upon admission for acute ischemic stroke. Crucially, this systemic anti-inflammatory profile was strongly associated with reduced neurological severity and favorable functional outcomes, with the most pronounced protective effects observed in patients with cerebral small-vessel disease (SVD). Such clinical findings provide compelling evidence that metformin's capacity to mitigate systemic – and potentially central – neuroinflammation translates into tangible preservation of neural and cognitive function in humans, serving as a critical bridge to understanding its microglia-mediated immunomodulation.

Microglia are key regulators of cerebral inflammatory homeostasis and synaptic pruning, and dysregulation of their autophagy–lysosomal pathway is considered a critical driver of exacerbated neuroinflammation in disorders such as AD and PD (Papini et al. 2024; Tzioras et al. 2023). Several animal studies have shown that in high-fat diet–induced or T2DM models, microglia display an "overactivated/senescent phenotype," accompanied by elevated reactive oxygen species (ROS), increased release of proinflammatory cytokines, and dysregulated synaptic phagocytosis, ultimately leading to impaired hippocampal neurogenesis and memory impairment (Chavda et al. 2024; González Pérez et al. 2025; Padhy et al. 2025).

Metformin can promote autophagic flux and lysosomal biogenesis by activating AMPK, inhibiting the mammalian target of rapamycin (mTOR), and regulating transcription factor EB (TFEB), thereby enhancing the cellular capacity for clearance of protein aggregates and damaged mitochondria (Campagnoli et al. 2025; Papini et al. 2024). In mouse models of insulin resistance–related dementia, metformin has been shown to restore a microglial "homeostatic" phenotype, reduce inflammatory mediators such as IL-1β, IL-6, and TNF-α, and improve synaptic structure as well as learning and memory performance (González Pérez et al. 2025). Expanding on this inflammatory mechanism, recent studies highlight that metformin targets the NLRP3 inflammasome via multiple convergent pathways. Hu et al. (2024) demonstrated in diabetic (db/db) mice that metformin activates SIRT1 to inhibit NLRP3 assembly, while Yang et al. (2019) revealed a parallel suppression via AMPK/mTOR-dependent mechanisms (Figure 2). These actions collectively block the maturation of proinflammatory cytokines and ameliorate cognitive dysfunction. Furthermore, this anti-inflammatory efficacy is reinforced by the upregulation of the Nrf2/HO-1 pathway, which suppresses oxidative stress and further dampens microglial activation (Xie et al. 2024). In T2DM-related cognitive impairment models, combined treatment with metformin and exercise synergistically attenuates hippocampal neuroinflammation, enhances adult neurogenesis, and significantly improves spatial memory (Padhy et al. 2025).

Beyond kinase inhibition, mechanistic studies reveal that metformin directly targets the protein phosphatase 2A (PP2A) complex, the major tau phosphatase. By interfering with the ubiquitin ligase MID1, metformin prevents the degradation of the PP2A catalytic subunit, thereby enhancing PP2A activity and promoting the dephosphorylation of tau protein at critical epitopes (Kickstein et al. 2010). Wnt/β-catenin signaling plays a critical role in synapse formation, adult hippocampal neurogenesis, and cognitive function, and its downregulation is closely associated with AD-related pathology (Padhy et al. 2025; Tzioras et al. 2023). In multiple AD and T2DM models, insulin resistance and inflammation can suppress Wnt signaling, leading to reduced synaptic protein expression and neuronal apoptosis (Kciuk et al. 2024; Madhusudhanan et al. 2020). Several preclinical studies suggest that metformin can indirectly modulate the Wnt/β-catenin pathway via AMPK and its downstream signaling, thereby improving synaptic structure and reducing tau hyperphosphorylation and oxidative stress (Campagnoli et al. 2025; de Mello et al. 2024; Kumar et al. 2023; Padhy et al. 2025). For example, in AD rat models, metformin or its nanoformulations, in combination with

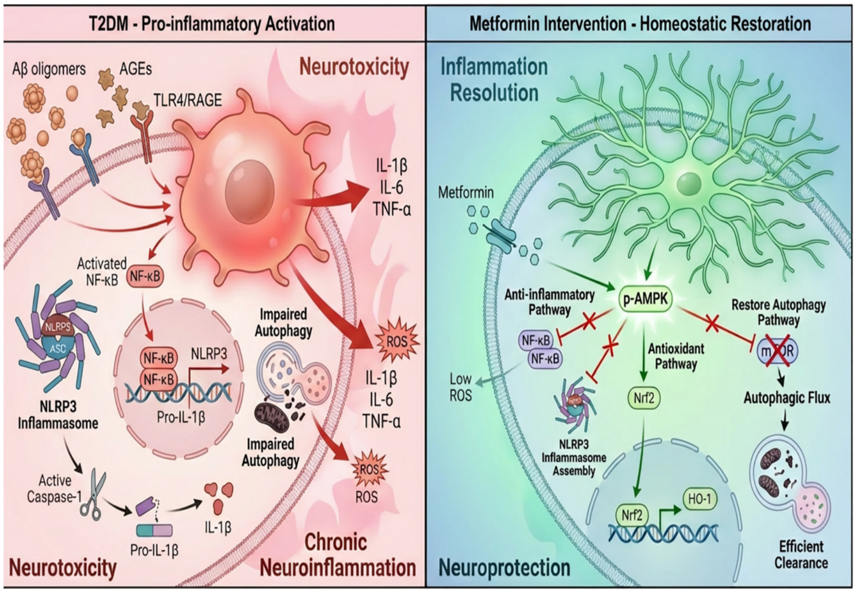

Figure 2: Metformin-mediated restoration of microglial homeostasis via AMPK-dependent pathways. The figure contrasts the proinflammatory state of microglia in type 2 diabetes mellitus (T2DM) with the homeostatic state following metformin intervention. Left panel (T2DM): pathological stimuli such as Aβ oligomers, advanced glycation end products (AGEs), and TLR4/RAGE signaling trigger the activation of NF-κB and the assembly of the NLRP3 inflammasome (comprising NLRP3, ASC, and pro-caspase-1). This results in the cleavage of pro-IL-1β into active IL-1β and the release of proinflammatory cytokines (IL-6, TNF-α), alongside impaired autophagy and increased reactive oxygen species (ROS), leading to neurotoxicity. Right panel (metformin intervention): metformin enters the cell and activates AMP-activated protein kinase (p-AMPK). Activated AMPK exerts neuroprotective effects by (1) inhibiting the NF-κB signaling and NLRP3 inflammasome assembly; (2) inhibiting mTOR to restore autophagic flux and promote the clearance of aggregates; and (3) activating the Nrf2/HO-1 antioxidant pathway to scavenge ROS. These mechanisms collectively resolve inflammation and restore microglial homeostasis.

other plant extracts, can significantly improve memory, lower phosphorylated tau levels, and partially restore AKT/ERK/GSK3β signaling (de Mello et al. 2024; Kumar et al. 2023). Parallel to Wnt signaling, the AMPK-CREB-BDNF axis serves as a crucial driver of synaptic maintenance. Metformin has been shown to protect against neuronal apoptosis and upregulate brain-derived neurotrophic factor (BDNF) through the phosphorylation of CREB (Liu et al. 2022). Furthermore, epigenetic mechanisms are involved; metformin enhances histone acetylation at the *BDNF* promoter region, facilitating CREB recruitment and the subsequent transcription of neurotrophic factors essential for restoring neurite plasticity (Fang et al. 2020). Finally, compelling evidence from nonhuman primates indicates that metformin can systemically attenuate biological aging. In a landmark 40-month study conducted in cynomolgus monkeys, metformin significantly slowed multi-tissue epigenetic aging trajectories – as measured by transcriptomic and DNA methylation–based aging clocks – with a particularly pronounced effect in the brain, where aging was reversed by an estimated 6 years. This neuroprotective effect was largely attributable to cell-autonomous activation of the Nrf2 antioxidant pathway, which robustly counter neuronal senescence. By targeting fundamental aging mechanisms at the epigenetic level, metformin offers the potential for a highly targeted therapeutic strategy to alleviate age-related cognitive impairment and metabolic-neural dysregulation, opening new avenues for exploration (Yang et al. 2024).

2.2.3 The "double-edged sword": mechanisms of potential risks

In many observational studies investigating metformin and dementia, insufficient adjustment for B12 deficiency and overall nutritional status may generate an apparent signal of "neurotoxicity" or mask metformin's true neuroprotective effects (Campbell et al. 2018; Luo et al. 2022). Thus, B12 status is a key confounder that must be considered when interpreting heterogeneous findings. Alvarez et al. (2025)

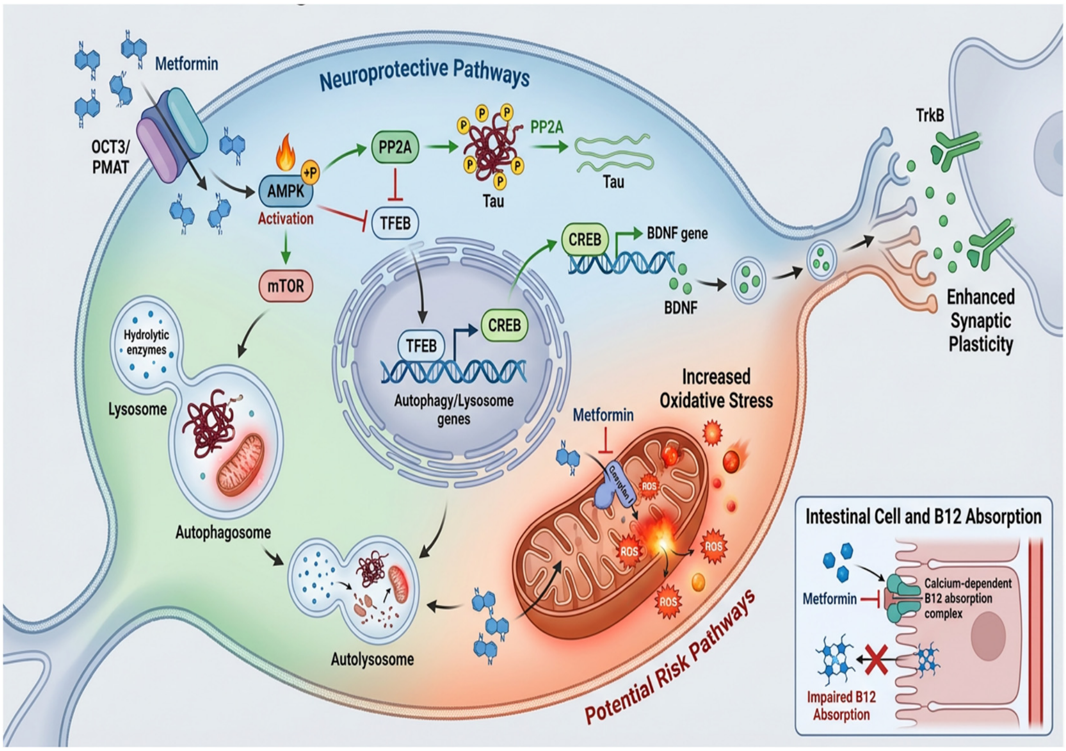

Figure 3: The "dual-edged sword" mechanism of metformin action on neurons. This diagram summarizes the balance between the direct neuroprotective pathways and potential risks associated with metformin treatment. Neuroprotective pathways (blue/green background): metformin enters neurons via organic cation transporters (OCT3/PMAT) and activates AMPK. This activation leads to (i) stimulation of PP2A activity, which dephosphorylates tau protein; (ii) regulation of the mTOR/TFEB axis to enhance lysosomal biogenesis and autophagy; and (iii) upregulation of CREB-mediated transcription of the *BDNF* gene, promoting BDNF release and TrkB signaling to enhance synaptic plasticity. Potential risk pathways (red background): conversely, metformin may inhibit mitochondrial complex I, potentially leading to increased oxidative stress (ROS) under conditions of metabolic vulnerability. Additionally, the inset highlights that metformin interferes with the calcium-dependent vitamin B12 absorption complex in intestinal cells, leading to impaired B12 absorption, a known risk factor for peripheral and central neuropathy.

reported that a serum B12 concentration below 148 pmol/L represents a clear deficiency threshold, while individuals with "borderline deficiency" (148–221 pmol/L) who receive long-term metformin therapy also show a significantly increased risk of cognitive impairment. Consequently, precise monitoring should not be limited to B12 concentrations alone but should also include homocysteine (Hcy) levels to detect earlier stages of functional B12 deficiency.

However, the inhibitory effects of metformin on mitochondria (for example, suppression of complex I activity) may, under certain conditions, exacerbate energy stress and ROS production, particularly in older individuals with preexisting mitochondrial dysfunction or in those receiving concomitant therapy with other mitochondria-toxic drugs (Campagnoli et al. 2025; Papini et al. 2024). Thus, the balance between "moderate stress" and "excessive stress" may determine whether metformin exerts protective or detrimental effects on the nervous system (Figure 3). Within a precision medicine framework, this "adverse effect" directly informs whether older adults, malnourished individuals, or patients with established diabetic peripheral neuropathy are suitable candidates for intensified metformin therapy.

3 Discussion

3.1 Based on the evidence summarized in this review, several relatively robust conclusions can be drawn

(1) T2DM substantially increases the risk of delirium and dementia, via multiple interacting pathways that include cerebral insulin resistance, mitochondrial dysfunction, chronic inflammation, and the cumulative burden of small-vessel disease (Komici et al. 2025; Madhusudhanan et al. 2020; Rebelos et al. 2021; Szablewski 2025);

(2) Compared with metformin, GLP-1RAs, SGLT2is, and TZDs have shown stronger and more consistent signals for dementia risk reduction across multiple studies, whereas DPP4is and conventional antidiabetic agents appear largely neutral (Hsieh et al. 2025; Hui et al. 2025; Sun et al. 2025c,d);
(3) MR studies indicate a positive causal relationship between T2DM and AD, whereas AMPK activation and other metformin-related targets are associated with protection in AD and stroke outcomes. This apparent discrepancy with some observational data largely reflects the influence of indication bias, dose and duration effects, and confounding by B12 deficiency (Geng et al. 2024; Liu et al. 2025; Wang et al. 2023);
(4) Mechanistically, metformin activates AMPK, enhances the autophagy–lysosome pathway, attenuates microglial inflammation, modulates mitochondrial ROS and the Wnt/β-catenin pathway, and may improve systemic metabolic–inflammatory status via the gut–brain and liver–brain axes, thereby offering multidimensional intervention potential against neurodegenerative pathology (Campagnoli et al. 2025; Jagust et al. 2023; Papini et al. 2024);
(5) However, the risk of B12 deficiency and peripheral/central neuropathy associated with long-term metformin use is a non-negligible adverse effect, requiring particular caution in older adults, malnourished individuals, and patients with established DPN (Campbell et al. 2018; Sayedali et al. 2023; Tang et al. 2024).
(6) Most longitudinal cohort studies and meta-analyses support that, among individuals with T2DM – particularly older adults and long-term adherent users – metformin is associated with a reduced risk of delirium and dementia/neurodegenerative disorders, although heterogeneous findings are seen in AD-specific populations and some Asian cohorts (Chin-Hsiao 2019; Luo et al. 2022; Sun et al. 2025a).

To explain the diametrically opposite effects of metformin on Alzheimer's disease risk observed between Asian and non-Asian populations, several specific factors and confounders must be considered, as suggested by recent literature:
(1) Genetics: The neuroprotective efficacy of metformin may heavily depend on specific genetic backgrounds, such as the APOE genotype. For instance, evidence suggests that metformin might be less effective in AD patients carrying the APOE ε4 allele, which varies across ethnicities (Luo et al. 2022).
(2) Diet: Dietary intake directly impacts baseline nutritional status. Metformin treatment has been linked to reduced levels of vitamin B12, a deficiency associated with cognitive impairment. Differences in regional diets may amplify the vulnerability to this metformin-induced B12 deficiency in certain populations (Luo et al. 2022).
(3) Comorbidity patterns: The presence of concurrent neurovascular conditions fundamentally alters metformin's cognitive trajectory. A recent clinical study demonstrated that metformin exerts significant neuroprotective effects and improves functional outcomes specifically in patients with concurrent cerebral small-vessel disease (SVD), suggesting that differing baseline comorbidity profiles among study cohorts can lead to conflicting results (Akiyama et al. 2024; Luo et al. 2022).
(4) Prescribing habits: Regional differences in prescribing behaviors significantly contribute to clinical heterogeneity. Specifically, varying exact thresholds of metformin exposure time, the duration of antidiabetic medication, and the choice between monotherapy versus combination therapies with other glucose-lowering agents remain major unadjusted confounders across different studies (Luo et al. 2022).

3.2 Clinical practice outlook: constructing a precision prescribing framework

3.2.1 Populations suitable for intensified metformin use

1) T2DM or high-risk individuals with concomitant metabolic syndrome/significant insulin resistance: these patients exhibit impaired insulin signaling both peripherally and centrally; by activating AMPK to improve metabolic status and inflammation, metformin could theoretically maximize its translation into neuroprotective effects (Campagnoli et al. 2025; Kciuk et al. 2024; Madhusudhanan et al. 2020; Rebelos et al. 2021).
2) Patients at the stage of early cognitive change (mild cognitive impairment or mild subjective cognitive decline): in early AD/MCI, metformin use in patients with comorbid diabetes is associated with a slower decline in global cognitive and functional scores compared with nonusers, with improvements in certain executive function measures (Koenig et al. 2017; Pomilio et al. 2022; Weinberg et al. 2024).
3) T2DM populations in whom microvascular complications predominate: both clinical and MR evidence support the role of metformin in improving endothelial function and reducing the burden of cerebral

small-vessel disease after stroke (Akiyama et al. 2024; Kim et al. 2024; Pakkam et al. 2024; Wang et al. 2023).

4) Individuals without evident B12 deficiency or prior DPN and with adequate nutritional status: in this population, the neuroprotective effects of metformin are less likely to be "offset" by B12 deficiency and malnutrition (Campbell et al. 2018; Sayedali et al. 2023; Zhang et al. 2022).

3.2.2 Populations in whom metformin should be used cautiously or avoided for intensified therapy

1) Very old adults (e.g., ≥80 years), and individuals who are frail or malnourished: in this population, B12 deficiency, anemia, and sarcopenia are more prevalent. Long-term metformin therapy may further deteriorate nutritional status via gastrointestinal adverse effects and B12 depletion, increase the risk of orthostatic hypotension and falls, and thereby indirectly impair cognition (Al-kuraishy et al. 2023; Campbell et al. 2018; Sayedali et al. 2023).
2) Patients with established diabetic peripheral neuropathy (DPN) or a history of B12 deficiency: in these patients, the severity of neuropathy and B12 reserves should be systematically evaluated. Where appropriate, the metformin dose should be cautiously reduced or alternative glucose-lowering regimens considered, alongside intensified B12 supplementation and close neurological follow-up (Campbell et al. 2018; Sayedali et al. 2023; Tang et al. 2024).
3) Patients with severe renal impairment or multiple organ failure: in this group, the risk of lactic acidosis and adverse drug reactions is heightened by polypharmacy (e.g., concomitant use of diuretics, RAS inhibitors, rifampin, etc.), and the central nervous system is likely to be in a "vulnerable state." Such patients are therefore unsuitable for metformin intensification as part of a "cognitive protection" strategy (Campagnoli et al. 2025; Papini et al. 2024).

3.2.3 Conceptual strategies for combination and sequenced use with SGLT2 inhibitors and GLP-1 receptor agonists

1) Combination optimization: assess the synergistic effects of metformin with other oral glucose-lowering agents and their impact on dementia risk, to better inform multidrug treatment strategies.
2) Combination strategy: in high-risk T2DM patients already on metformin, preferentially add a GLP-1 RA or SGLT2i to achieve integrated cardiovascular, renal, and neuroprotective benefits.
3) Sequential strategy: for individuals at high cognitive risk but constrained by the cost or tolerability of GLP-1 RAs, a stepwise approach starting with metformin and gradually introducing more potent agents may represent an effective intervention option.

3.3 Future research directions

3.3.1 Key considerations in study design

1) Outcome selection: endpoints should be diversified to include acute and chronic outcomes as well as biomarkers, encompassing in-hospital delirium, risk of incident MCI, and relevant neuroimaging features.
2) Population and stratification: the study population should be clearly defined with respect to age, comorbidities, and cognitive risk factors, to ensure broad generalizability of the findings.
3) Intervention and comparison groups: study designs should incorporate metformin versus placebo, as well as head-to-head comparisons with other glucose-lowering agents, to identify the optimal therapeutic strategy.
4) Integration of MR and drug-target genetics: a causal framework linking drug targets to clinical outcomes should be constructed by combining MR with target-based genetics, thereby enhancing the biological interpretability of the evidence.

3.3.2 Mechanistic and clinical translation research

1) Thoroughly elucidate the synergistic mechanisms of dual therapy combining metformin with novel glucose-lowering agents;
2) Conduct high-quality randomized controlled trials to validate the clinical effectiveness of combination and sequential strategies for delirium and dementia;
3) Develop personalized treatment models guided by biomarkers and genetic risk profiles.

Building on this, future work should prioritize metformin as a core component of a "metabolic–brain protection" strategy in patients with coexisting metabolic syndrome/significant insulin resistance, early MCI, or T2DM primarily characterized by microvascular complications. Newer agents such as GLP-1 RAs and SGLT2is should be introduced at clinically appropriate time points to achieve integrated, multiorgan benefits for the heart, kidneys, and brain. In parallel, high-quality RCTs and MR studies should incorporate delirium, MCI, and both imaging and fluid biomarkers as key endpoints, with the ultimate goal of establishing a genuine "precision pharmacotherapy–precision prevention" framework that slows the progression of T2DM patients from acute brain dysfunction to long-term dementia.

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